# Response to the Editor and Reviewer 1

Manuscript QSS-2026-0077 (Major Revision)

Citation-Network Cohesion Across Low- and High-Impact Journals: A Matched Author Analysis

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Dear Dr. Larivière,

Thank you for inviting a major revision of manuscript QSS-2026-0077. We also thank Reviewer 1 for the detailed assessment and for identifying the need for a clearer and more rigorously justified contribution. We rebuilt the affected analysis before revising the prose. The reviewer comments are reproduced below in their original order, followed by point-by-point responses. Page and line references correspond to the final compiled revision.

## Comment 1: Overall assessment

> *This paper investigates whether authors publishing in low-impact journals exhibit denser and more reciprocated citation patterns compared to their counterparts in high-impact venues. The topic is of undoubted interest, but in its current form, it prioritises technical virtuosity over substantive contribution. For a journal like QSS, which values methodological rigour alongside clarity, a major revision is essential. The authors must reframe their work for a broader audience, clarify their analytical pipeline, and strengthen the connection between their findings and the scholarly literature. However, given the number and severity of the issues identified below, I regret to recommend rejection in its current form.*

**Response.** We agree that the earlier draft required a substantive rather than cosmetic revision. We retitled the manuscript from *Citation Cliques in Low Impact Journals* to *Citation-Network Cohesion Across Low- and High-Impact Journals: A Matched Author Analysis* to reflect its associational matched design and remove the unsupported implication of established citation cliques. The paper is now framed as an associational matched comparison, not an assessment of scientific merit or evidence of intent. We retain the motivating directional comparison as an explicit hypothesis about higher primary cohesion and outgoing concentration among Cases, while distinguishing positive mean differences from zero-inflated median shifts and avoiding any claim about intent. The analytical sequence is shown in a reader-facing design figure and presented as Data and Cohort, Matching, Network Construction and Definitions, Paired Inference, Anomaly Screen, Results, Discussion, Limitations, and Conclusion. The feature set and claims were reduced, affected results were regenerated from one canonical workflow, and the Discussion now interprets the findings in relation to the author- and journal-level literature. The revision reports weaker, null, or contrary results rather than retaining claims from the earlier analysis.

**Location.** Title, Abstract, Figure 1, Discussion, and Conclusion; pp. 1–16 and 22–25; revised manuscript lines 1–50, 74–280, and 320–417.

## Comment 2: Clarity and Presentation: terminology

> *The primary obstacle is the paper's prose. Specialised terminology is deployed without adequate explanation, and crucial concepts are often introduced before being defined. This disrupts the narrative flow and assumes a level of technical familiarity that may not align with the journal's readership. I suggest to the authors an effort to translate the methodological procedures into clear language.*

**Response.** We rewrote the Methods so that the unit of analysis, data identifiers, matching variable, graph populations, and notation are defined before the analyses that use them. The revision expands DOI, ISSN, and HHI at first use; defines $h_5$ operationally; distinguishes the outgoing ego, matched-author, and work-reference populations; and gives both formal and plain-language definitions of the directed graph and its binary projection. It also explains Isolation Forest as repeated random partitioning, defines the direction of its study score, explains weighted betweenness and assortativity, and states how positive effect sizes should be read. The main text now gives the reader-facing explanation first, while the appendix retains the exact implementation details needed for reproducibility.

**Location.** Data and Cohort through Anomaly Screen; pp. 4–16; Table 2; revised manuscript lines 74–279.

## Comment 3: Clarity and Presentation: metric definitions and burst intensity

> *The presentation of metrics in Table 5 exemplifies this problem. Several measures lack clear, self-contained definitions, and the bibliographic references provided are not illuminating: they point to general theoretical frameworks without specifying how these translate into the precise operationalisations actually adopted in the study. The Normalised burst intensity, for instance, is defined as* $\max_j w_{ij}/(\text{in\_strength}+1)$ *and described as the largest single-peer citation weight normalised by total incoming volume, yet the notation (*$w_{ij}$*, `in_strength`) and the connection to Kleinberg's burst-detection methods remain under-explained. What is missing is an explicit link between the inspiring principle and its concrete implementation, which for a journal like QSS should be made transparent.*

**Response.** The former Table 5 has been replaced by Table 2. It now states each formula, denominator, analytic population, and interpretation. Its notes define every symbol used in the table, explain the HHI scale, distinguish structural zeros from unavailable denominators, and specify missing-value handling. The surrounding Methods also gives the exact reference-deduplication and fractional-weight construction that supplies those formulas.

The former normalised burst-intensity variable has been removed. It is replaced by the study-defined maximum annual dyadic surge share, whose complete formula compares consecutive dyad–years and divides the largest positive increment by the author's total identified outgoing weight. Absent dyad–years are treated as zero for this calculation, authors without identified outgoing weight remain missing, and the result is bounded by $[0, 1]$. The manuscript explicitly states that this is a five-year scalar summary, not a trajectory model or an implementation of Kleinberg's state-transition burst method; the Kleinberg attribution has therefore been removed. References now indicate conceptual provenance; each study-specific operationalization is stated in full rather than delegated to those sources.

**Location.** Network Construction and Definitions, Paired Inference, and Limitations; pp. 7–13 and 25; Table 2; revised manuscript lines 145–196 and 398–401.

## Comment 4: Clarity and Presentation: organization, K-means, and literature

> *Furthermore, the paper conflates methods, analysis, and results, often presenting findings before the underlying techniques have been introduced. The three central groups from the k-means clustering, for instance, appear without any description of the clustering procedure. Restructuring the narrative to follow a logical sequence would greatly improve comprehension. Importantly, the discussion fails to connect the empirical results back to the literature review, making the paper feel like an isolated technical exercise rather than a contribution to the ongoing scholarly conversation.*

**Response.** All essential definitions and analytical decisions now precede Results, which begins only after the Data, Matching, Network, Paired Inference, and Anomaly Screen sections. Figure 1 summarizes that sequence. K-means and the former three-cluster result have been removed because they are no longer part of the revised study.

The Discussion now connects the regenerated findings, not only the study design, to prior work. It distinguishes outgoing HHI from Evdaimon et al.'s incoming-citer concentration indicator, uses the tie-heavy zero-median findings to limit interpretation of cohesion differences, contrasts lower journal endogamy and higher surge share with Heneberg's journal-level kinetics, and explains why the enrichment and mixing results do not reproduce the excessive-edge group detection studied by Kojaku et al. It also discusses non-adversarial alternatives and states why these studies are conceptually informative but not numerically comparable.

**Location.** Methods through Results, and Discussion; pp. 4–16 and 22–24; revised manuscript lines 74–280 and 320–377.

## Comment 5: Methodological Rigour and Transparency: matching

> *The technical execution, while ambitious, contains critical problems. The matching methodology suffers from a logical inconsistency: the bucketing formula $b = \lfloor h_5/3 \rfloor$ permits a five-point $h_5$-index difference, contradicting the intended three-point limit. Additionally, no sorting order is specified for the greedy one-pass algorithm. These fundamental issues must be resolved to ensure methodological transparency and credibility.*

**Response.** We replaced the bucket rule with the explicit inclusive eligibility condition $|h_{5,\mathrm{Case}} - h_{5,\mathrm{Control}}| \leq 3$. The revised Methods defines $h_5$ and specifies the complete greedy order: subject, ascending absolute $h_5$ difference, Case ORCID, and Control ORCID. Selection is without replacement within subject. We also disclose a retained upstream restriction that the earlier draft omitted: the $h_5$ table materializes only positive values, so otherwise tier-eligible zero-$h_5$ profiles do not enter candidate generation.

An immutable read-only audit recomputed all candidates and the greedy traversal from the source author profiles and materialized positive $h_5$ values. Within that implemented pool it reproduced the retained 9,431-pair order, membership, and pair-set hash exactly. It does not claim to reproduce a rematch that admits $h_5 = 0$. The maximum observed absolute difference is 2, with 0 caliper violations and no reused author–subject keys. Matching balance is reported by subject, and the four primary comparisons are repeated for exact-$h_5$ pairs.

**Location.** Matching and sensitivity results; pp. 6–7 and 18–19; Tables 1 and 5; revised manuscript lines 115–142 and 295–302.

## Comment 6: Methodological Rigour and Transparency: outlier framework

> *The outlier-detection framework also requires substantial justification. The rationale for feature selection, hand-tuned weighting, and the four-sigma cutoff is not adequately explained. The use of feature weights in an Isolation Forest model is mathematically redundant, as the algorithm is not designed for such inputs, and the four-sigma threshold implies a normality assumption that is inappropriate for skewed bibliometric data. The authors should provide a clearer justification for these decisions, report sensitivity analyses to test the stability of their findings, and clarify whether the Isolation Forest and Cohesion Composite Score capture redundant information.*

**Response.** The revised specification uses an explicit structural rule rather than an optimized feature search. It retains five author-edge measures corresponding to self-reference, prior-collaborator citation, mutual exchange, neighborhood closure, and recipient concentration. Journal endogamy is excluded because it is a work-reference measure, balance because it is a signed internal-flow position unavailable without internal incident weight, and surge share because it is temporal. We present this as a transparent revision choice, not as an originally prespecified or independently validated optimum.

No additional feature weights are applied. The four-sigma rule and former Cohesion Composite Score have been removed from the revised canonical analysis and manuscript. Controls define subject-specific median/IQR scaling, model fits, and score cutoffs; a nonfinite or zero Control IQR is replaced by 1 solely to avoid division by zero. The declared 99th Control percentile is a stringent empirical-tail operating point that makes no Gaussian assumption and is not claimed to be uniquely optimal. The library's `contamination=auto` decision cutoff is not used; the reported criterion is computed directly from negated `score_samples`.

Sensitivity analyses evaluate 97.5%, 99%, and 99.5% Control percentiles across ten fixed seeds and report leave-one-feature-out fits. Because all four cohesion-restriction measures also enter the detector, the manuscript explicitly treats their intersection as an overlapping guardrail rather than independent corroboration. The complete overlap table and rank association quantify that dependence. Among 3,870 screenable author–subject rows, 200 exceeded the detector threshold, 55 met the cohesion restriction, and 53 met both; 147 were detector-only and 2 restriction-only. Detector score and cohesion exceedance count had Spearman $\rho = 0.385$ ($p < 0.001$). Across 5 leave-one-feature-out detector fits, final-flag Jaccard similarity with the canonical flags ranged from 0.907 to 0.981, and Case/Control enrichment ranged from 15.15 to 16.80.

**Location.** Anomaly Screen and anomaly results; pp. 14–16, 19–20, and 31; Tables 6, 7, 8, and 10; revised manuscript lines 218–279 and 304–312.

## Comment 7: Inconsistencies and Omissions

> *Several inconsistencies and omissions further weaken the analysis. The manuscript does not specify the underlying statistical test applied to the matched pairs, a critical omission given the non-normal nature of the data. The claim of a widening gap in "citation trajectories" is purely descriptive and no time series data are reported to support this assertion. On page 9, the authors stated that "normalised burst intensity" was dropped, leaving 13 features for the analysis. However, on page 13, they report that "12 out of 14 tested behavioral metrics remain significant". The authors should clarify whether the missing feature was reintegrated for the univariate statistical tests (by dropping the incomplete pairs) or if this is a typographical error.*

**Response.** The revised Methods specifies a two-sided paired Wilcoxon signed-rank test for every outcome because several paired-difference distributions are bounded and tie-heavy. Zero differences are omitted from the signed ranks, unavailable pairs are deleted separately for each outcome, and Wilcoxon $p$-values are Benjamini–Hochberg adjusted within the four-outcome primary and secondary families. Each table reports its own complete-pair count, median paired difference, mean paired difference, both paired-bootstrap intervals, matched rank-biserial effect, and sign-flip randomization check. Because the measures are zero-inflated, mean paired differences and their intervals are descriptive complements; they do not replace the median-based nonparametric summary. All four primary mean paired differences favor Cases, while three median differences are zero because many pairs are tied.

We could not verify from the retained legacy materials whether the earlier 13-versus-14 discrepancy arose because normalised burst intensity was reintegrated for per-metric tests or because one count was typographical. We therefore withdraw both legacy statements rather than assigning them a retrospective explanation. The rebuilt analysis reports eight outcomes in two declared families, separately from five screen inputs. The former burst variable is replaced by maximum annual dyadic surge share, with 6,908 complete pairs in its secondary comparison.

We removed the earlier widening-gap claim. The year field supports coauthor timing and the scalar surge-share construction, but the revision estimates neither tier-specific trajectories nor a tier-by-year interaction. The Limitations now states this explicitly.

**Location.** Paired Inference, matched results, and Limitations; pp. 13, 16–19, and 25; Tables 3, 4, and 5; revised manuscript lines 193–215, 285–302, and 398–401.

## Comment 8: Network Characterisation

> *The central qualitative claim of a "hub-and-spoke" topology is presented without any formal network definition, graph-theoretic metrics, or quantitative thresholds. The role of Louvain community detection is also unclear. From Table 5 it is unclear if the adjustements for directed networks are adopted in the computation of newtork indicators auch as eigenfactor. Moreover, it is unclear on which graphs some indicators are calculated (in table 5 the manuscript refers to "undirected projection" and "undirected citation" graphs without specifications. To move from observation to verifiable evidence, the authors must provide the analytical basis for their network characterisation.*

**Response.** The former hub-and-spoke characterization, qualitative topology labels, Louvain community detection, and modularity result have been removed from the revised canonical analysis and manuscript.

No community or topology result is now inferred.

The revision separates the journal and author networks. Eigenfactor is a fixed journal-level exposure used before author matching, not an author centrality. The Methods now specifies citing-to-cited orientation, journal self-citation removal, outgoing-column normalization, 0.85 damping, article-count redistribution of dangling mass and teleportation, and within-subject normalization. The current workflow consumes the fixed database scores; because the intermediate journal-edge and article-count exports are unavailable, the manuscript explicitly states that the revision cannot independently reproduce or verify the upstream scores or extraction window end to end.

For the author analysis, the Methods formally defines the loopless directed weighted graph $G_s$ and its binary undirected projection $U_s$. It identifies the population, direction, weight, denominator, and missingness rule for every measure. Direction is discarded only for $U_s$ clustering; reciprocity, balance, and tier mixing retain fractional weights and direction. The label-swap $p$-value tests the within-tier fractional-weight share, while weighted assortativity is reported separately as a descriptive statistic. No community or topology result is inferred.

**Location.** Data and Cohort, Network Construction, Anomaly Screen, and network results; pp. 4–5, 7–11, 15–16, and 20–21; Tables 2 and 9; revised manuscript lines 83–104, 145–190, 261–279, and 314–319.

## Comment 9: Conclusion

> *In summary, this manuscript contains interesting ideas, but the priority for revision must be clarity, rigorous methodological justification, and a stronger connection to the literature. The current draft reads as a proof of technical capability. I suggest that the authors transform it into an accessible contribution to quantitative science studies.*

**Response.** We agree with this priority. The revised paper leads with the substantive matched comparison, defines its technical terms and populations before use, limits the analysis to auditable operationalizations, reports uncertainty and sensitivity rather than a proliferation of exploratory outputs, and interprets the actual regenerated findings against the relevant literature. The primary tables now show both median and mean paired differences, making the zero-inflated, tie-heavy outcome distributions explicit. Its conclusion is deliberately associational: the screen can prioritize descriptive follow-up but cannot establish coordination, intent, or misconduct.

**Location.** Abstract, Discussion, Limitations, and Conclusion; pp. 1–2 and 22–25; revised manuscript lines 1–18 and 320–417.

## Additional corrections incorporated in the revision

Beyond the issues named above, the rebuild makes subject part of every author and edge key; deduplicates work references before author expansion; counts complete author teams in fractional denominators while requiring identified graph endpoints; separates ego, induced-graph, and work-reference populations; preserves unavailable values; and uses the same final flag for enrichment and figures. All manuscript values, tables, and figures are generated by one declared canonical analysis; earlier estimates are not used.

**Location.** Author–subject cohort through Paired Inference, Anomaly Screen, Results, and Data and Code Availability; pp. 5–16 and 26; revised manuscript lines 105–215, 250–251, 280–284, and 428–438.